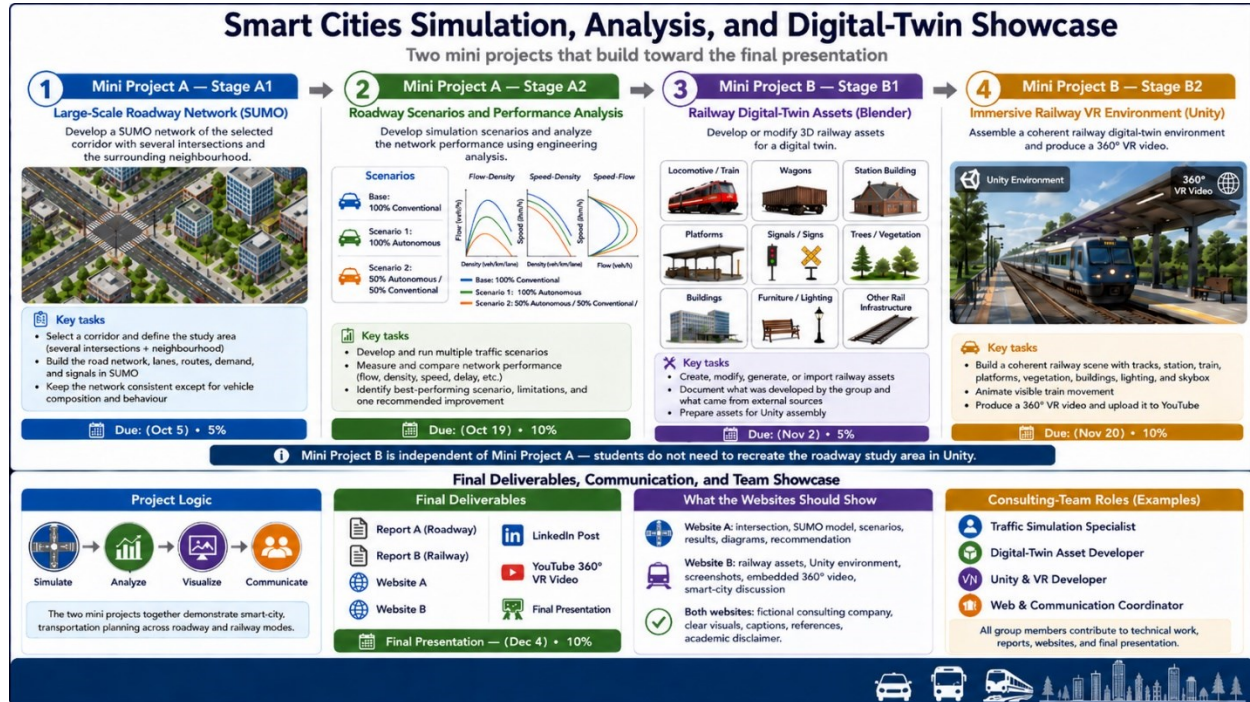


Mini Projects — SMRTTECH 4SC3

Smart Cities Simulation, Analysis, and Digital-Twin Showcase



Sample Technical Report:

<https://youtu.be/DsGvDcmw5ds>

<https://youtu.be/21tJPYt951s>

Web-Based Project Showcase:

<https://sepidch10043.github.io/>

YouTube Link:

https://youtu.be/-t2MWsJf_7o

LinkedIn Post:

Explained in this file.

Project Overview

Students will work in groups of three or four and act as a smart transportation consulting team. Each group will create a fictional consulting-company name and will complete two separate client projects during the term.

Mini Project A - Roadway Traffic Simulation and Analysis (SUMO)

Mini Project B - Railway Digital Twin and Immersive Visualization (Blender and Unity)

The two mini projects are independent engagements. Mini Project B does not require students to recreate the roadway study area from Mini Project A. Together, the two projects demonstrate how simulation, engineering analysis, and digital twins can support smart-city transportation planning across both roadway and railway modes.

Each mini project has its own staged submissions, its own technical report, and its own project website. A report template will be provided for each mini project.

Deliverables and Deadlines

Deliverable	Content	Due	Weight
Mini Project A - Stage 1	SUMO network of the selected intersection	Oct 5	5%
Mini Project A - Stage 2	Performance analysis + Report A	Oct 19	10%
Mini Project B - Stage 1	Railway digital-twin assets (Blender)	Nov 2	5%
Mini Project B - Stage 2	Unity VR environment + Report B	Nov 20	10%
Final Presentation	Two project websites, LinkedIn post, presentation	Dec 4	10%

Late submissions are subject to the course late policy. All group members are expected to contribute to the technical work, engineering analysis, reports, websites, and final presentation.

Mini Project A - Roadway Traffic Simulation and Analysis

Client brief: Autonomous vehicles are expected to play an increasingly important role in urban transportation over the next decade. As cities prepare for this transition, they need to understand how different levels of autonomous-vehicle adoption may affect traffic flow, speed, density, congestion, and overall intersection performance.

Your consulting team has been hired by a fictional city to select a congested major intersection, develop three traffic scenarios in SUMO, and evaluate the effects of conventional, autonomous, and mixed traffic conditions.

Objectives of Mini Project A:

- Develop simulated urban areas (intersections and road segments) with conventional and autonomous vehicles
- Analyze roadway performance using traffic flow, density, speed, and fundamental diagrams
- Recommend an operational improvement supported by the simulation results

Stage A1 - Urban Roadway Simulation in SUMO

Assigned Study Area

Each group has been assigned a study area in the City of Toronto. Use the GIS map provided for your group as the reference for developing your SUMO road network.

The study area includes signalized intersections, major roads, and secondary and local roads, providing sufficient roadway complexity for the project.

SUMO Network Development

Each group must develop a SUMO model that reasonably represents the selected study area. You should reasonably apply Millstones 1 & 2 & 3 in Lab 1. The network should include:

- Roadway and junction geometry that reasonably matches the provided GIS map.
- Roadway lanes and turning lanes, with appropriate lane numbers and layouts.
- Correct lane-to-lane connections and turning movements at all junctions-not only at major intersections.
- All major, minor, secondary, local, and neighbourhood roads, as well as any highways visible within the assigned GIS map.

Do not limit your model to one intersection or only the major roads. All roads and junctions within the map must be represented, except for the exclusions below.

- Parking spaces and driveways do not need to be modelled.

The objective is to develop a complete road network whose geometry, lane layout, and connections match the provided map as closely as reasonably possible.

Stage A1 focuses on road-network development. Traffic signal implementation, traffic flow, right-of-way settings, speed limits, and scenario development will be addressed in Stage A2.

Submission files:

1. Create a folder named with your group number, for example, Group1.
2. Include the following files, **using the standard SUMO file extensions shown below. Eclipse SUMO**

Required file	Example filename
SUMO network file	Group1.net.xml
NETEDIT configuration file	Group1.netecfg
SUMO simulation configuration file	Group1.sumocfg

3. Do not include the GIS image in your submission.
4. Compress your group folder into a single ZIP file, such as Group1.zip, and have one group member submit it through Avenue on behalf of the group.

Provided Files and Map Setup:

Before developing your road network, read your group's TXT file. It contains:

- Map width (W) and height (H): Enter these values in Edit Visualization when setting up your GIS background map.
- Real-world location coordinates: These are provided for geographic reference, to help you identify the assigned study area, connections, etc.

Use the GIS image and setup values provided specifically for your group. Set up the background map before beginning your road-network development.